Terminal-Bench testset - construction summary

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1. Pipeline (one line)

241 TB tasks -> Qwen3-Embedding-0.6B + FAISS top-5 over 44,787 ClawHub skills -> Opus 4.7 LLM-as-judge per (task, skill) pair -> YES verdicts kept as gt_skills -> 62 validated tasks pushed as testset.

2. Key metrics - judge run

Judge model	claude-opus-4-7 (Max-plan via 'claude -p')
(task, skill) pairs judged	1205 = 241 tasks x top-5
YES verdicts	98 (8.1 %)
NO verdicts	1106 (91.8 %)
Unparseable (AUP refusal)	1 (treated as NO)
Wall-clock	~25 min (concurrency = 4)
Marginal cost	$0 (Max plan)

3. Key metrics - validated testset

Tasks kept (>= 1 YES)	62 / 241 (25.7 %)
Tasks dropped (all NO)	179 (74.3 %)
Single-GT tasks	40
Multi-GT tasks (>= 2 YES)	22 (2: 12, 3: 7, 4: 2, 5: 1)
YES distribution by FAISS rank	r1: 33 r2: 21 r3: 14 r4: 18 r5: 12
Output file	data/terminal_bench_validated.jsonl
Schema	{task_id, task_description, gt_skills:[{skill_id, slug, name, rank, reason}], n_yes,

Todo completion vs. spec

Original spec (paraphrased)

(2) Build TB testset by retrieving top-k ClawHub skills via Qwen embedding, incrementally inject each skill into the agent's skills library, run the agent on the task, treat skills that improve agent performance as GT. Drop tasks where no skill helps. Push metadata to GitHub.

(3) Build a test script that mimics how harbor delivers a task to the agent: feed task description, require the agent to emit selected skills in <skill> format, parse output.

(4) Vary distractors: pool sizes {5, 50, 500} x noise {random, hard, easy}.

(5) Analyse model outputs, compute pick accuracy, log results.

Status by point

Spec point	Status	Note
(2a) Top-k retrieval from ClawHub	**DONE**	Qwen3-Embedding-0.6B + FAISS over 44,787 skills; top-5 stored in terminal_bench_topk.jsonl.
(2b) Per-skill helpfulness signal	**SUBSTITUTED**	Originally: incremental injection via harbor. Replaced by Opus LLM-as-judge after harbor was ruled out (no Max-plan support, $700-$1.7K API cost, conflicted with retrieval-isolation guardrail). User approved 2026-05-04.
(2c) Drop tasks with no helpful skill	**DONE**	179/241 dropped (74.3 %). Kept set: 62 tasks in terminal_bench_validated.jsonl.
(2d) Push metadata to GitHub	**PENDING**	Commit prepared (testsets/data/* + skill_selection_eval/* + RESULTS.md update). Awaiting user confirmation on push target / branch.
(3a) Read harbor's task-delivery code	**DONE**	harbor_terminus_2_skills.py::_build_skill_prompt_xml + _handle_skill_tool_calls_xml documented in testsets/README.md.
(3b) Mimic harbor prompt format	**DONE**	skill_selection_eval/prompt.py reproduces <available_skills> XML block + skill-system instructions verbatim.
(3c) Parse <skill> output	**DONE**	skill_selection_eval/parser.py uses identical regex to harbor's _handle_skill_tool_calls_xml.
(3d) Wire validated GT into eval	**DONE**	run_tb.py now surfaces validated_gt_slugs from terminal_bench_validated.jsonl in trial output.
(4a) Pool sizes 5 / 50 / 500	**DONE**	tb_pool_builder.py implements all three; pilot run hit them on 30 tasks (2026-05-03-tb-pilot.jsonl).
(4b) Noise modes random / hard / easy	**DONE**	Implemented + topk_only + none variants. Same module.
(4c) Run grid on validated 62-task set	**DONE**	930 trials, 0 errors. Initial conc=6 hit Max-plan rate-limit at 747; resumed conc=4. Total ~80 min.
(5) Pick-accuracy analysis on validated GT	**DONE**	Multi-GT scoring (Hit@1, Recall, MRR) per (size, noise) cell. Selection-collapse pattern reproduces (sz=5 hard 69.4 % -> sz=500 hard 41.9 %).

Validated-grid eval (points 4c/5)

930 trials = 62 tasks x {5, 50, 500} pool sizes x {random, hard, easy, topk_only, none} noise. Backend: claude-sonnet-4-5 via 'claude -p' Max plan. 0 errors. Multi-GT scoring: Hit@1 = top-1 in GT set; Recall = |picks intersect GT| / |GT|; MRR over picked-list.

size	noise	n	Hit@1	Recall	MRR	refusal	in_topk
5	random	62	74.2%	58.1%	0.742	12.9%	87.1%
5	hard	62	69.4%	54.1%	0.694	17.7%	82.3%
5	easy	62	75.8%	58.9%	0.758	17.7%	82.3%
5	topk_only	62	74.2%	57.6%	0.742	16.1%	83.9%
5	none	62	-	-	-	100.0%	0.0%
50	random	62	75.8%	59.5%	0.758	12.9%	85.5%
50	hard	62	54.8%	45.1%	0.548	19.4%	59.7%
50	easy	62	75.8%	57.0%	0.758	17.7%	82.3%
50	topk_only	62	80.6%	64.3%	0.806	11.3%	88.7%
50	none	62	-	-	-	96.8%	0.0%
500	random	62	58.1%	42.8%	0.581	30.6%	62.9%
500	hard	62	41.9%	34.2%	0.419	27.4%	43.5%
500	easy	62	48.4%	36.2%	0.484	40.3%	56.5%
500	topk_only	62	79.0%	63.5%	0.798	17.7%	82.3%
500	none	62	-	-	-	83.9%	0.0%

Headline findings

1. Validated GT recovers a real signal: Hit@1 in small / topk_only settings (74-81 %) matches SkillsBench's GT-aware grid. The Opus-judge GT is a meaningful target.

2. Selection collapse reproduces on TB. Hard-noise Hit@1 falls 69.4 % (sz=5) -> 54.8 % (sz=50) -> 41.9 % (sz=500): same shape as SkillsBench (59.6 -> 41.6). The phenomenon is now cross-dataset.

3. topk_only is the stability ceiling: at sz=500 (79.0 %) it matches sz=5 (74.2 %). Failure mode is distractors mixed near GT, not pool size.

4. No-GT refusal degrades with size (100 % -> 96.8 % -> 83.9 %). Identical signature to SkillsBench. 16 % sz=500 hallucination rate is a real safety concern.

Caveats & next steps

Methodology caveats

- Single judge model (Opus 4.7). Two-judge agreement (e.g. Opus + GPT) would give precision/recall-leaning splits and is the standard for paper-grade GT.

- Strict prompt may over-reject niche-but-valid skills. 8.1 % YES rate may be a lower bound; relaxing to 'plausibly helpful' would lift it.

- Judge sees only skill name + ~200-char description (the corpus only stores summaries). Full-text SKILL.md retrieval would likely raise YES rate.

- 1 unparseable judgement was an Anthropic AUP refusal on a content-sensitive task description; treated as NO.

- Single seed across both judge and (planned) eval grid - no variance estimate yet.

What's left to close points (4c) and (5)

```
# 1. Re-run TB grid on validated 62-task set
python -m testsets.skill_selection_eval.run_tb \
  --tasks testsets/data/terminal_bench_validated.jsonl \
  --out   testsets/runs/2026-05-05-tb-validated-grid.jsonl \
  --pool-sizes 5,50,500 --noise-modes random,hard,easy,topk_only,none \
  --seeds 0 --conc 6 --model claude-sonnet-4-5

# 2. Aggregate Hit@1 / Recall / MRR / refusal vs validated_gt_slugs
python -m testsets.skill_selection_eval.aggregate_tb \
  --in  testsets/runs/2026-05-05-tb-validated-grid.jsonl \
  --out testsets/runs/2026-05-05-tb-validated-grid.summary.md
```

Files produced this session

```
testsets/skill_selection_eval/judge_tb.py         # NEW Opus-judge driver
testsets/skill_selection_eval/aggregate_judge.py # NEW judge -> validated.jsonl
testsets/skill_selection_eval/run_tb.py          # updated: surfaces validated_gt_slugs
testsets/runs/2026-05-04-tb-judge.jsonl          # 1205 judgements (raw)
testsets/data/terminal_bench_validated.jsonl     # 62-task GT testset
testsets/data/terminal_bench_validated_summary.md
testsets/RESULTS.md                              # +new section
```